The Strategic Game

A Signaling Game of Espionage and Deception

The Game

- Two intelligence agencies — **MI6** and **Cheka** — are in pursuit of retrieving a very important secret file.
- The file is hidden in one of three cities: **Prague**, **Lapland**, or **Geneva**. The mission is to find the file, and both agencies gain points accordingly.

Points Distribution

Outcome	**MI6**	**Cheka**
Only MI6 finds the file	10	0
Only Cheka finds the file	0	10
Both find the file	3	3
Neither finds the file	0	0

Notice that both agencies would rather find the file *alone* than share credit — the reward for exclusivity (10) is far greater than the reward for a shared success (3). This is what makes deception valuable.

What Happens in Each Mission?

1) One of the three cities is chosen at random, and the file is hidden there.

2) An intelligence report is received by MI6 from an unknown source.

- The report says one thing: *"The file is hidden in ____ city."*
- But MI6 is not sure whether this information is 100% correct or not.
- The report gives the **correct** city with probability r.

Example: Suppose the report says the file is in Geneva. Then

$$P(\text{report is true}) = r$$

$$P(\text{true location} = \text{Geneva}) = r$$
$$P(\text{true location} = \text{Prague}) = \frac{1-r}{2}$$
$$P(\text{true location} = \text{Lapland}) = \frac{1-r}{2}$$

3) After MI6 receives the report, it does two things:

a) **Sends a message** to Cheka (through a middle man) about the file's location.

b) **Searches** for the file, using the report as a guide.

- The middle man gives Cheka the location *exactly as stated in the report* with probability p_t *(the probability of truth-telling)*.
- MI6 follows the report and searches the city given in the report with probability p_f *(the probability of following)*.

4) Before Cheka receives the message, the probability of searching any given city was equal ($= \frac{1}{3}$ each). But after receiving the message, Cheka updates its belief via **Bayesian updating**, since it knows the probability p_t with which the middle man relays the true location.

Disclaimer

- We will see how, even if the middle man could be giving *wrong* information to Cheka, it would still help Cheka in finding the city (Bayesian update!).
- We will also see how bluffing is *not always* the best option.

After Cheka updates its belief, the city with the **maximum posterior probability** is chosen for the search.

Now we repeat this mission many times, and see who wins and gains the most points.